

hw11

Please make sure you're following all the instructions described in the "Homework Instructions" page on Canvas (go to the homepage of our class and you'll see a link saying "Homework Instructions" just underneath the header of the page).

Note on Homework 11:

Autograding isn't particularly conducive to plots. As such, the only questions that will make use of the autograder are those that produce some output that the autograder can check. The other questions will show up as having a "out of" score of 0 - these will be graded manually.

Also, for *all* plots, provide useful titles and axis labels.

1 Insect Sprays

Adapted from "The Book of R" by Davies, Exercise 14.1

The `InsectSprays` dataset contains the counts of an insect, the tobacco hornworm, in agricultural experimental units treated with six different insecticides (see `?InsectSprays` for more details).

1.1 *Histogram*

Produce a histogram of the counts of insects.

No autograder for this question.

1.2 *Total number of insects*

Obtain the total number of insects found according to each spray. Then, produce a vertical barplot of these totals, labeling each plot appropriately.

var: `tot` - numeric vector; the total number of insects for each spray

1.3 *Boxplots*

Generate boxplots of the counts of insects according to each spray type. Add a gray, dashed horizontal line representing the overall mean. Also include appropriate axis labels and a title.

No autograder for this question.

2 Earthquakes

Adapted from “The Book of R” by Davies - Exercise 14.1i-k

For this question we’ll be using the built-in `quakes` dataset, which provides of 1000 seismic events near Fiji since 1964 (see `?quakes` for more details).

2.1 *Magnitude vs stations detecting*

The dataset contains information on both the magnitude of the earthquake and the number of stations reporting. Use a scatter plot to explore the relationship between these two variables. In addition, add a fitted line. You can use the following code to find the slope and intercept of the least squares regression line:

```
coefs <- lm(mag ~ stations, quakes)$coefficients
```

This returns a two-element vector with the intercept and slope for the line. Use these coefficients with `abline()` to plot the trend line (you’ll want to take a look at the help file for the function). Make use of the graphical parameters to make the plot look nice.

Note: For the line to be added to the plot, `mag` must be on the y axis and `stations` must be on the x axis.

No autograder for this question.

2.2 Classify the earthquakes

First, create a factor corresponding to the magnitudes (do you remember which function we can use to classify a numeric vector)? Each entry should assume one of three categories - the break between the first and second categories should be the $\frac{1}{3}$ quantile and the break between the second and third categories should be the $\frac{2}{3}$ quantile. The labels for the categories should be “low”, “medium”, and “high”.

Note: The `quantile()` function can be used to find the quantiles - check the documentation and/or do some web searching to learn how to use it.

var: `quakes2`: a copy of `quakes` that has an additional column called `category`, created as described above.

2.3 Plot the earthquakes

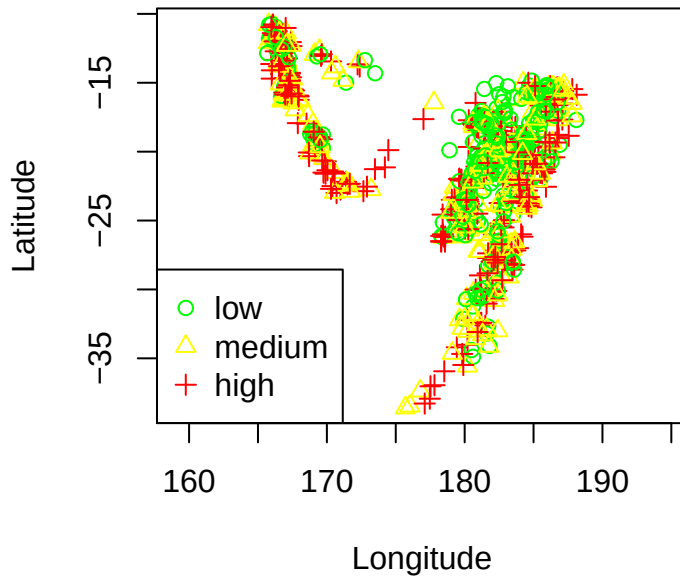
Re-create the plot shown next, where low-, medium-, and high-magnitude events are plotted with `pch` being assigned 1, 2, and 3 and `col` being assigned "green", "yellow", and "red", respectively. Add a legend to the plot indicating which shape corresponds with each category.

Note that while there are more efficient ways of doing this, you may want to just break the data into three subsets based on the category and then plot each category separately.

Note: you can use `asp=1` in the call to `plot()` to ensure that spacing on the x- and y-axes are equal.

No autograder for this question.

Earthquakes near Fiji



3 Histogram of all numeric columns

Using the `penguins` data, write code to plot histograms of all of the numeric columns. Identify the numeric columns programmatically - don't type out the column names. Create a 2 row, 3 column grid containing the 5 plots (there will be one blank space). You'll want to make use of some stuff from Lesson 6 (applying functions to columns) as well as Lesson 10 (for loops).

No autograder for this question.